

FREEDOM INTERNATIONAL SCHOOL

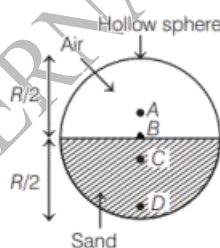
WORKSHEET-MCQ

PHYSICS

CLASS XI

SYSTEM OF PARTICLES AND ROTATIONAL MOTION

- The centre of mass of a system of particles does not depend on
 - masses of the particles
 - internal forces of the particles
 - position of the particles
 - relative distance between two particles
- Two bodies of masses 1 kg and 2 kg are lying on the x-y plane at (1, 2) and (-1, 3) respectively. What are the coordinates of the centre of mass?
 - (2, -1)
 - $(\frac{8}{3}, \frac{-1}{3})$
 - $(\frac{-1}{3}, \frac{8}{3})$
 - (-1, 2)
- A ball kept in a closed box moves in the box making collisions with the walls. The box is kept on a smooth surface. The velocity of the centre of mass
 - of the box remains constant
 - of the box and the ball system remain constant
 - of the ball remains constant
 - of the ball relative to the box remains constant
- Which of the following points is the likely position of the centre of mass of the system shown in figure?



- A
 - B
 - C
 - D
- Three identical spheres each of mass M are placed at the corners of an equilateral triangle of side 2m. Taking one of the corners as the origin, the position vector of the centre of mass is
 - $\sqrt{3}(\hat{i} - \hat{j})$
 - $\frac{\hat{i}}{\sqrt{3}} + \hat{j}$
 - $\frac{\hat{i} + \hat{j}}{3}$
 - $(\hat{i} + \frac{\hat{j}}{\sqrt{3}})$
 - Newton's second law for rotational motion of a system of a particle can be represented as (L for a system of particles)
 - $\frac{dp}{dt} = \tau_{\text{ext}}$
 - $\frac{dL}{dt} = \tau_{\text{int}}$
 - $\frac{dL}{dt} = \tau_{\text{ext}}$
 - $\frac{dL}{dt} = \tau_{\text{ext}} + \tau_{\text{int}}$
 - A disc of mass M and radius R is rotating about one of its diameters. The value of radius of gyration for the disc is



- R/4
 - R/2
 - R/6
 - R/8
- By keeping the moment of inertia of a body constant, if we double the time period, then the angular momentum of the body

- (a) remains constant (b) becomes half (c) doubles (d) quadruples
9. Two bodies have their moments of inertia I and $2I$ respectively about their axis of rotation. If their kinetic energies of rotation are equal, their angular momenta will be in the ratio
(a) 1:2 (b) $\sqrt{2}:1$ (c) 2:1 (d) $1:\sqrt{2}$
10. A body having a moment of inertia about its axis of rotation equal to 3 kg m^2 is rotating with angular velocity of 3 rad s^{-1} . Kinetic energy of this rotating body is same as that of a body of mass 27 kg moving with a velocity v . The value of v is
(a) 1 ms^{-1} (b) 0.5 ms^{-1} (c) 2 ms^{-1} (d) 1.5 ms^{-1}
11. An isolated particle of mass m is moving in a horizontal plane (x - y) along the x axis at a certain height above the ground. It suddenly explodes into two fragments of masses $m/4$ and $3m/4$. An instant later, the smaller fragment is at $y = +15 \text{ cm}$. The larger fragment at this instant is at
(a) $y = -5 \text{ cm}$ (b) $y = +20 \text{ cm}$ (c) $y = +5 \text{ cm}$ (d) $y = -20 \text{ cm}$